LAB-III TASK 5 lab Report

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Section B

Title:

Implementation of Else-if Ladder in C

Objective:

To classify a number as positive, negative, or zero using the else-if ladder.

Theory:

The `else-if` ladder is used when multiple conditions need to be checked sequentially. It allows the program to choose one block of code among many alternatives.

Syntax:

```
if (condition1)
{
    // block 1
}
else if (condition2)
{
    // block 2
}
else
{
    // default block
}
```

Only the first true condition is executed, and the rest are skipped.

Program (with comments):

```
#include <stdio.h>   // Input-output library

int main()
{
    int num; // Declare variable

    printf("Enter an integer: "); // Prompt user
    scanf("%d", &num);            // Store input

    if (num > 0)                  // Check if number is positive
    {
        printf("The number is positive.\n");
    }
    else if (num < 0)             // Check if number is negative
    {
        printf("The number is negative.\n");
    }
    else                          // If neither positive nor negative
    {
        printf("The number is zero.\n");
    }

    return 0; // End program
}
```

Detailed Explanation:

The program takes an integer input from the user.

It then evaluates multiple conditions:

1. `num > 0` → checks if the number is positive
2. `num < 0` → checks if the number is negative
3. If neither condition is true, the number must be zero

The program executes only one block:

- If the first condition is true → prints positive
- Else if second condition is true → prints negative
- Otherwise → prints zero

This structure ensures efficient evaluation of multiple conditions.

Output (Example):

Enter an integer: -5
The number is negative.

Conclusion:

The else-if ladder allows multiple conditions to be tested efficiently and is essential for complex decision-making in programming.